

Main
+ start(Stage): void
+ main(String[]): void
+ Main():

+ 1
+ 1
+ 1

MatrixOperations
+ copy(int[][]): int[][]
+ merge(int[], int[], int, int): int[]
+ checkRemoving(int[]): ClearRow
+ intersect(int[], int[], int, int): boolean
- checkOutOfBounds(int[], int, int): boolean
+ detectFullRows(int[]): int[]
+ deepCopyList(List<int[]>): List<int[]>
- MatrixOperations():

NotificationPanel
+ showScore(ObservableList<Node>): void
+ NotificationPanel(String):

<<interface>> BrickGenerator
brick: Brick
nextBrick: Brick

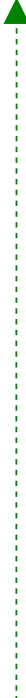
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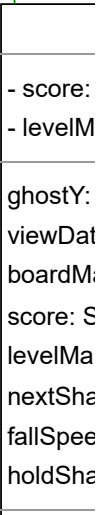
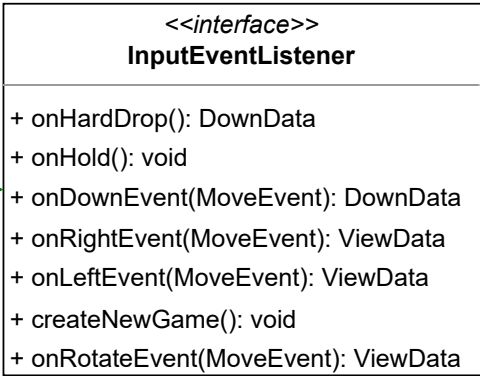
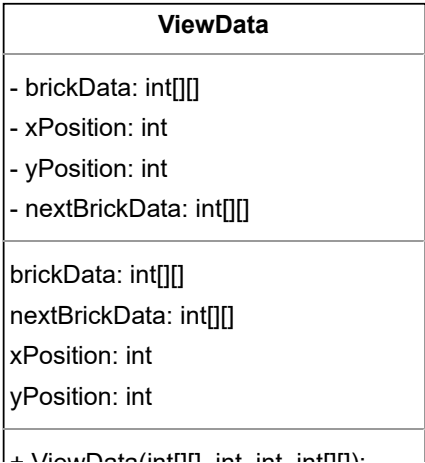
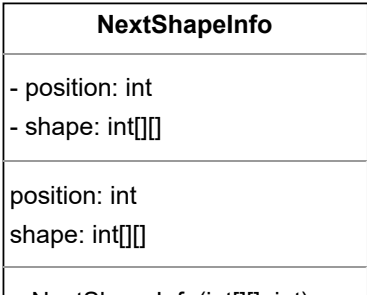
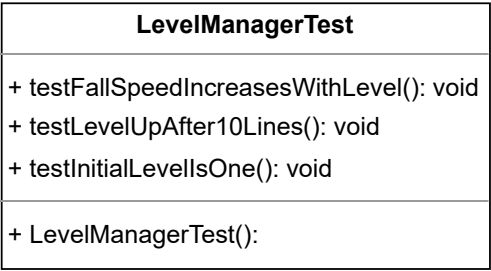
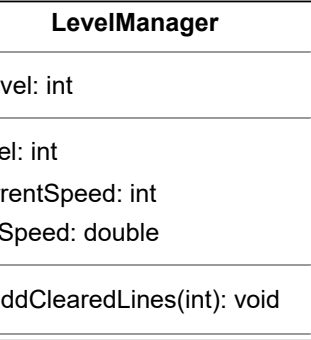
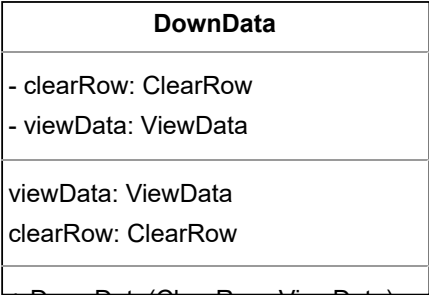
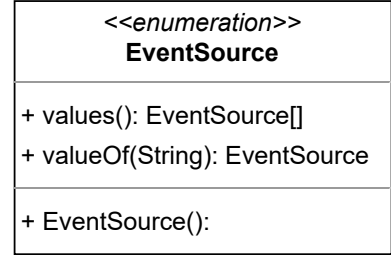
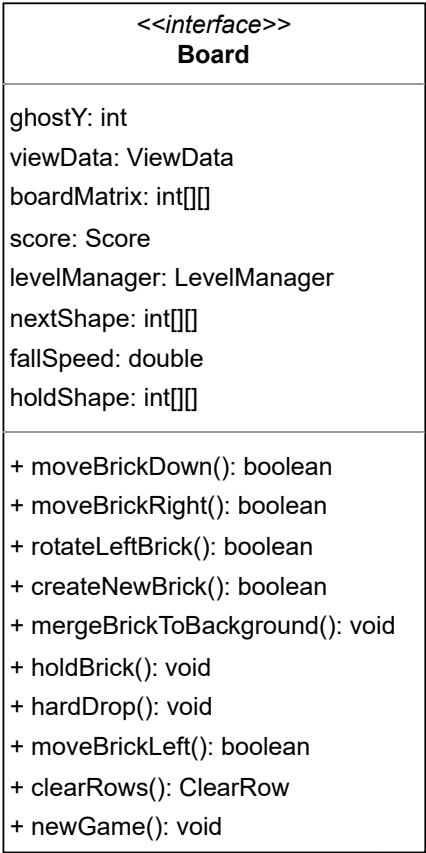
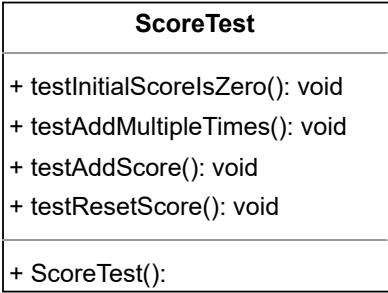
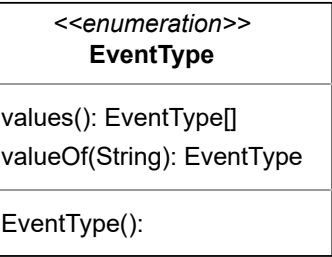
ClearRow
- scoreBonus: int
- rows: int[]
- linesRemoved: int
- newMatrix: int[][]
newMatrix: int[][]
linesRemoved: int
rows: int[]
scoreBonus: int
+ ClearRow(int, int[], int, int):

GameOverPanel
- restartButton: Button
restartButton: Button
+ GameOverPanel():

RandomBrickGenerator
brick: Brick
nextBrick: Brick
+ RandomBrickGenerator():

BrickRotator
- currentShape: int
- brick: Brick
currentShape: int[]
nextShape: NextShapeInfo
brick: Brick





SimpleBoard
Score anager: LevelManager
int ta: ViewData atrix: int[][] Score anager: LevelManager ape: int[][] ed: double ape: int[][]

DICK. DICK
+ BrickDetector()

GuiController
- eventListener: InputEventListener
eventListener: InputEventListener
+ hideBrick(): void + updateCombo(int): void + flashRows(int[], boolean): void + onPauseClicked(): void + updateLinesCleared(int): void + onRestartClicked(): void + initialize(URL, ResourceBundle): void + newGame(ActionEvent): void + bindScore(IntegerProperty): void + initGameView(int[[]], ViewData): void - getFillColor(int): Paint - setRectangleData(int, Rectangle): void - moveDown(MoveEvent): void + refreshHold(int[[]]): void + refreshGhost(ViewData, int): void + showFloatingScore(int): void + onStartClicked(): void + updateLevel(int): void + refreshBrick(ViewData): void - togglePause(): void + refreshNext(int[[]]): void + refreshGameBackground(int[[]]): void + gameOver(): void - initBGM(): void + onQuitClicked(): void
+ GuiController():

MoveEvent
- eventType: EventType
- eventSource: EventSource
eventType: EventType
eventSource: EventSource
MoveEvent(EventSource, EventType)

SimpleBoardClearRowTest
+ testClearSingleFullRow(): void
+ testClearMultipleRows(): void
+ SimpleBoardClearRowTest():

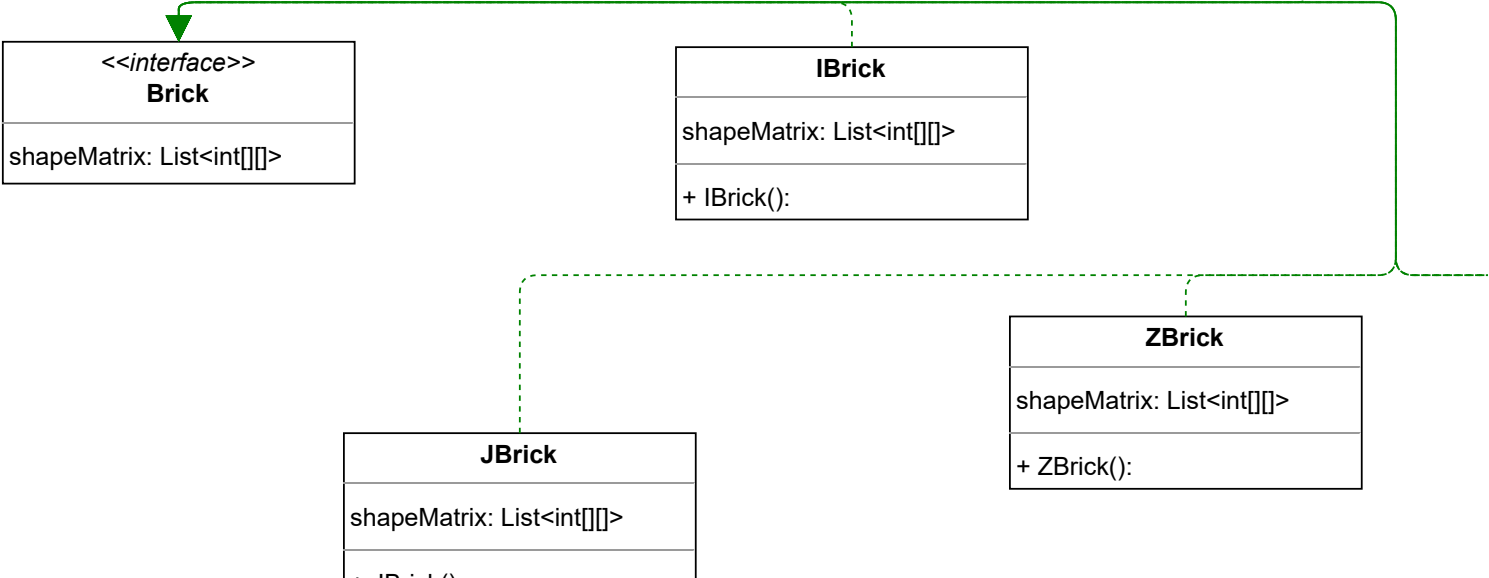
TBrick
shapeMatrix: List<int[[]]>
+ TBrick():

<<interface>>
Brick
shapeMatrix: List<int[[]]>

IBrick
shapeMatrix: List<int[[]]>
+ IBrick():

JBrick
shapeMatrix: List<int[[]]>
+ JBrick():

ZBrick
shapeMatrix: List<int[[]]>
+ ZBrick():



+ View/ data(int[][] int int int[][])



Score
+ scoreProperty(): IntegerProperty + add(int): void + reset(): void
+ Score():

GameController
- board: Board
board: Board
+ onHold(): void + onHardDrop(): DownData + onRightEvent(MoveEvent): ViewData + onDownEvent(MoveEvent): DownData + onLeftEvent(MoveEvent): ViewData + createNewGame(): void + onRotateEvent(MoveEvent): ViewData - playClearAnimation(int[], Runnable): void

MatrixOperationsTest
+ testDetectMultipleFullRows(): void + testIntersect(): void + testMerge(): void + testDetectFullRows(): void
+ MatrixOperationsTest():

SimpleBoardHoldTest
+ testFirstHoldStoresCurrentBrick(): void + testCannotHoldTwiceInSameTurn(): void + testHoldResetsAfterSpawnNewBrick(): void
+ SimpleBoardHoldTest():

SimpleBoardMovementTest
+ testBrickCanMoveDownAtStart(): void + testMoveLeft(): void + testBrickStopsWhenTouchingBottom(): void + testMoveRight(): void
+ SimpleBoardMovementTest():

- + holdB
- + moveI
- + clearF
- + moveI
- + moveI
- + rotate
- + create
- + newG
- + merge
- tryMov
- + hardD
- + Simple



OBrick
shapeMatrix: List<int[][]>
+ OBrick():

SBrick
shapeMatrix: List<int[][]>
+ SBrick():

LBrick
shapeMatrix: List<int[][]>
+ LBrick():



rick(): void
BrickRight(): boolean
Rows(): ClearRow
BrickLeft(): boolean
BrickDown(): boolean
LeftBrick(): boolean
NewBrick(): boolean
Game(): void
BrickToBackground(): void
MoveDownWithoutRender(): boolean
Drop(): void
GameBoard(int, int):

+ JBrick():

